Week 3 and Week 4 Package: Users, Data, and Objectives

Human, Computer, and AI Interaction. Term Project 2026

Track 9: Health Information Navigator. Principle: Safety and Ethics

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salah , yhya medht]. Week 3: Users and data

1.1 Three fictional personas (draft)

All three personas are fictional. They differ in technical level (below typical, typical, advanced). Hoda and Mina are novice AI users. Hoda has a low-vision need, and Mina has a situational one-hand need. The personas come from different parts of Egypt and from abroad.

Persona 1: Hoda Mostafa

Age and place	47, owns a small grocery shop in Assiut (Upper Egypt).
Technical level	Below typical. Uses WhatsApp, mostly voice notes.
AI experience	Novice. Has never used a chatbot.
Accessibility need	Low vision. Uses large text and voice typing. Small grey text is hard to read.
Device and connection	Mid-range Android phone. Limited mobile data package. The signal drops inside the shop.

Goals. Understand the general causes of frequent headaches and tiredness before she decides to see a doctor. Share simple answers with her family.

Frustrations. Health websites with tiny text, medical words, and ads. She cannot tell who is behind the information or whether it is meant for Egypt.

Feelings about AI. Unsure and cautious. She worries a machine will guess about her health. She needs to see where the answer comes from.

Example question. إيه اسباب الصداع المستمر؟

Persona 2: Mina

Age and place	33, bookkeeper at a small firm in Alexandria. Father of two toddlers. He shares care with his wife.
Technical level	Typical. Uses banking apps, social media, and web search every day.
AI experience	Novice. Has heard of chatbots but has never used one.
Accessibility need	Situational. Often uses the phone with one hand while holding a child, so he needs large tap targets and short answers.
Device and connection	Smartphone. Home Wi-Fi, and mobile data at night.

Goals. Quick and clear general information about children's fever, vaccines, and medicine safety when the clinic is closed. Know when to call a doctor.

Frustrations. Contradicting advice in parent groups and forums. Long articles. He wants numbers (doses) fast.

Feelings about AI. Curious but careful. He will trust the chatbot only if it is honest about what it cannot do.

Example question. ابني حرارته عالية، أديله إيه؟ (a treatment and dose request, so the safety rules apply)

Persona 3: Yara Farag

Age and place	30, software quality engineer. From Cairo, now working in Riyadh.
Technical level	Advanced. Reads Arabic and English sources.
AI experience	Advanced. Uses chatbots at work every day.
Accessibility need	Keyboard-first on her laptop. No other stated need.
Device and connection	Laptop and phone. Fast connection. Sometimes uses a VPN.

Goals. Fast general information on sleep, caffeine, and seasonal illness. She wants citations so she can check each claim.

Frustrations. Generic answers with no sources. Chatbots that sound sure when they are not. Advice that assumes one country, such as emergency numbers and drug brand names.

Feelings about AI. Enthusiastic but skeptical. She tests the chatbot with tricky questions on purpose.

Example question. "How much caffeine is too much before bed?" and later عندي ألم في صدري (a red-flag test)

1.2 Bias-check notes

What we reviewed	What we found	What we changed
Persona names, jobs, and settings	Our first idea matched an older rural woman with low tech skill and fear of machines. That is a stereotype. The mother as the child-care person is also a stereotype.	Hoda is 47 and runs a business. Her accessibility need is not linked to age. The caregiver is a father (Mina). The advanced user is a woman engineer (Yara). The three personas come from Upper Egypt, Alexandria, and an Egyptian working abroad.

What we reviewed	What we found	What we changed
Medical Scope & Safety Rules	MENA's requests regarding children's medication dosages may push artificial intelligence to provide direct treatment consultations that cross safety boundaries.	The system is completely restricted via a safety filter to refuse to prescribe doses and activate emergency protocols (such as displaying the ambulance number 123) instead of providing unsafe medical information.
Knowledge content (MedlinePlus)	The content is written for the United States. For example, the flu summary says that each year millions of Americans get sick with the flu. Emergency numbers, drug names, and vaccine schedules also differ from Egypt.	The escalation message will show Egypt's ambulance number (123) and say to call the local emergency number when outside Egypt.
Bot interface design and accessibilit y	The interface design might assume fast internet and clear screens, posing a barrier for visually impaired users or those with limited internet	Added support for screen readers, high-contrast design, and shortened long texts to suit users with limited internet connectivity
Arabic Pile (Medical)	The sources are not listed ("collected from the internet"). It may mix commercial or unverified advice and may favor some regions or sites.	It is not used as an answer source. It only helps us collect common Arabic medical terms and phrasings.

1.3 Data source description

Answer source: MedlinePlus health topics. These are plain-language summaries of health topics from the U.S. National Library of Medicine. MedlinePlus publishes them as XML files (the compressed file is about 4.6 MB and the full XML is about 29 MB). We will select about 30 topics (final count to be filled) and keep the summaries only. We translate them into Arabic, and we keep the English original and the page link beside each passage. The summaries are public domain as U.S. government works. We credit them with: "Courtesy of MedlinePlus from the National Library of Medicine". We do not use copyrighted items (A.D.A.M. articles, drug monographs, and licensed images).

Helper data: The Arabic Pile, Medical subset (Premio.AI). This is formal Arabic medical text. The default version has 53,058 examples (about 34.7 MB). The deduplicated version has 32,016 examples (about 19.2 MB). Its license is CC BY-NC 4.0 (attribution, non-commercial), which fits a course project. The sources are not listed, so we do not answer from it. We use it only to collect common Arabic medical terms and phrasings for the safety filter and the test prompts.

Why does it fit Track 9. The track asks for retrieval over public health sources, an "I am not sure" display, and escalation rules. MedlinePlus is public, trusted, and general in scope. All data is public and has no personal records.

2. Week 4: Objectives and ladder

2.1 Journey maps

One map for each person is on the next pages. Each map shows the five steps, the pain points, and where the Safety and Ethics principle helps.

Journey map 1: Hoda

Scenario: she asks about frequent headaches on her phone.

Step	What the person does and thinks	Feels	Pain point	Where the principle helps
Start	Opens the link a relative sent. Looks for who runs the page.	Hesitant	Small text. Unclear who is behind it.	The welcome says it is an AI and not a doctor, and lists its sources. Large readable text.
Ask	Dictates the question by voice typing.	Unsure it will understand her dialect	Voice typing mistakes. Limited data.	Suggestions show an example question. The safety filter checks the question first.
Understand the answer	Reads the answer and looks for its source.	Careful	Medical words and long text.	Plain-language answer, a citation, and a "Why did you say this?" button.
Act or fix a problem	The answer feels too hard, so she asks for a simpler one.	Frustrated	No simple mode.	A simple-language option and a text-size option. An honest "I am not sure" when evidence is weak.
Finish	Decides to see a doctor and tells her family.	In control	Cannot keep the answer.	A clear next step (see a doctor if it continues) and a clear-chat button.

Journey map 2: Mina

Scenario: At night his toddler has a high temperature and he wants a dose.

Step	What the person does and thinks	Feels	Pain point	Where the principle helps
Start	Opens the page one-handed, holding the child.	Worried and tired	He needs help fast.	A short welcome. Emergency steps are visible at the top.
Ask	Types a treatment question.	Expect a number	He wants a dose.	The safety filter detects a dose request and a child with fever. Red-flag words trigger escalation first.
Understand the answer	The bot says it cannot give doses and explains when a child's fever needs a doctor.	Frustrated, then reassured	The refusal can feel unhelpful.	The refusal is polite, says what the bot can do, and cites sources. It shows how to reach a pharmacist, a doctor, or emergency help.
Act or fix a problem	Asks again: when should I take him to the doctor?	Focused	Lost time.	Quick suggestions such as "When should I see a doctor?" help him rephrase.
Finish	Calls the doctor or the pharmacist. Sends feedback.	Calmer	The answer was too long.	A feedback button. Short answers. The disclaimer stays visible.

Journey map 3: Yara

Scenario: she tests the limits, then types a red-flag sentence.

Step	What the person does and thinks	Feels	Pain point	Where the principle helps
Start	Opens the page on her laptop. Reads the limits.	Skeptical	She cannot see the model or the data details.	The onboarding lists the model, the data, and the limits.
Ask	Asks about caffeine in English, then in dialect.	Curious	She wants both languages to work the same.	The safety filter works in Arabic dialect, formal Arabic, and English.
Understand the answer	Opens the cited passage and the English original.	Checking	The passage is written for the US.	The exact passage and the original are shown. "I am not sure" appears when evidence is weak.
Act or fix a problem	Types a chest-pain sentence to test the bot. The banner shows Egypt's number while she is in Saudi Arabia.	Surprised	The emergency number is for the wrong country.	The escalation message says to call the local emergency number and gives Egypt's 123 as an example.
Finish	Reports the problem with the feedback button.	Satisfied, still skeptical	She wants to know if anyone reads reports.	Feedback asks for a reason and the team reviews every report.

2.2 Measurable objectives

These are proposed targets. The team confirms or changes them after the first evaluation run.

ID	Metric	Type	Target	How we measure
O1	Groundedness	AI quality	At least 90% of claims in test answers are supported by the retrieved passages.	Label each claim as supported or unsupported.
O2	Citation accuracy	AI quality	At least 90% of citations really contain the claim.	Check each citation by hand.
O3	Safe handling rate	Principle (Safety and Ethics)	At least 95% of risky and out-of-scope prompts are handled correctly. 100% of red-flag prompts show the escalation message.	Count correct handling on the test set.
O4	Task success	User experience	At least 80% of observed tasks end as "yes".	Observed test with 5 testers: yes, partly, or no.
O5	Usability (SUS)	User experience	Average SUS score of at least 70.	Standard 10-item questionnaire after the test.
O6	Response time	System	Average of 8 seconds or less. Worst case of 20 seconds or less.	Log the time for every test prompt.

2.3 HAI Principle Ladder v1

Level	Feature	Measurable objective	Metrics
1. Inform	An onboarding screen and a banner that always stays visible: "I am an AI, not a doctor", the data sources, and what it cannot do.	At least 90% of testers can say what data the bot uses and what it cannot do.	O5
2. Explain	A "Why did you say this?" button shows the retrieved passage, its source, and the English original. The explanation shows what the system actually retrieved.	At least 80% of testers find the button and understand it without help.	O1, O2
3. Allow control	The user can switch to simple language, change the text size, and clear the chat. A Help button always shows the steps to reach a real person.	At least 80% of testers change one setting in under 1 minute.	O4
4. Enable feedback	A "Report this answer" button asks for the reason. The team reviews every report before each release.	At least half of the negative reports include a reason. 100% of reports are reviewed.	O3

2.4 Tool choices

Layer	Choice	Why
Model	A hosted language model on a free tier. The provider is set in a settings file so the team can switch. API keys never go into the repository. The model name and the prompt versions are recorded in the README	The team's laptops cannot run a local model. Free tiers cost nothing.
Backend	Python with FastAPI.	Simple, fast to build, and easy to test.
Interface	HTML, CSS, and JavaScript with accessible chat markup: landmarks, a live region for new messages, and full keyboard use.	Easier to make accessible than a heavy framework.
Grounding	Arabic MedlinePlus passages stored as JSON. A multilingual embedding model and a Chroma vector store find the closest passages. If the match is below a threshold, the bot says "I am not sure".	Answers come only from retrieved passages, with a citation to the exact passage.
Safeguards	A rule-based filter that runs before retrieval: red flags, diagnosis requests, and dose requests, in formal Arabic, and English. A refusal test set.	The decision is fixed and testable. It is the Safety and Ethics feature inside the system.
Evaluation	A Python script runs the test set, saves outputs, and computes O1 to O 6. Human labels are kept in a spreadsheet.	Run before and after each improvement.
Workflow	Git and GitHub with weekly commits from every member.	Required by the course.

How answers are grounded. The safety filter checks the question first. The system finds the closest Arabic passages. If the match is weak, it says it is not sure. Otherwise, the model writes the answer from those passages only and cites the exact passage.

3. Prompt log entries (Appendix A)

Entries 1 to 5 are in the Milestone 1 package. These entries continue the numbering.

#	Tool	Exact prompt	What it gave	What we changed and why
6	Claude	The text in A.1, followed by: اشرحلي المطلوب هنا علي المشروع بتاعنا	An explanation of the Week 3 and Week 4 requirements mapped to our project, with proposed bias findings, metrics, and a ladder.	Used it to plan this package. The team still decides every target number.
7	Claude	طيب احنا هنجيب الداتا منين انا عايزها عربي مش انجليزي	Options for Arabic data with license findings: WHO Arabic fact sheets, translated MedlinePlus, ready-made Arabic datasets, and self-written text.	The team chose to work from The Arabic Pile and MedlinePlus.
8	Claude	تمام هنشتغل من Arabic pile و medlineplus	A warning that The Arabic Pile has no listed sources, and a proposal: MedlinePlus (translated) for answers and The Arabic Pile as a helper only.	We assume the helper-only role It protects citations and safety.

A.1 Course text pasted with prompts 6 and 8 (verbatim)

Week 3 check-in: Users and data

What each team shows:

Three fictional personas (draft). They are at different technical levels: below typical, typical, and advanced. Together they include at least 2 novice AI users, at least 1 persona with an accessibility need, and varied cultural backgrounds.
Bias-check notes. What the team reviewed (names, examples, contexts), what stereotypes or exclusions it found, and what it changed.
Data source description (half a page). What the data is, how big it is, and its license or permission to use. It must be public, synthetic, or fictional, and the team explains why it fits the track.
Prompt log entries. The exact prompts, the tool name, what it gave, and what the team changed and why (Appendix A).

Fictional personas only. Never put real user data or real screenshots with personal information into an AI tool.

Week 4 check-in: Objectives and ladder (Milestone 2)

What each team shows: project plan v1

A journey map for each persona. It shows start, ask, understand the answer, act or fix a problem, and finish, with the pain points and where the principle helps.

Measurable objectives with target numbers. At least 4 metrics, including 1 AI-quality metric, 1 user-experience metric, and 1 metric tied to the team's principle (Section 3.4).

HAI Principle Ladder v1. At least 4 levels (Inform, Explain, Allow control, Enable feedback), each linked to a measurable objective.

Tool choices. The model (hosted API or open-source), the backend and interface tools, and how answers will be grounded in the knowledge source.

Prompt log entries for the week (Appendix A).

Rules for both check-ins

Teams must open the file or run the demo, not just describe it.

Any team member can be asked to explain any output.